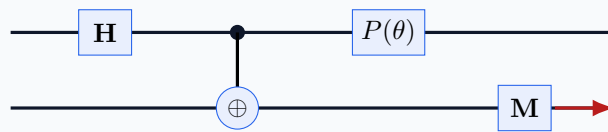


# Classical and Quantum Theory for QPU

From Bits to Bloch Spheres



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A ground-up explanation of classical logic, complex amplitudes, quantum state, measurement, entanglement, and the mathematics represented by QPU gates.

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# 1 Classical computation foundations

Quantum circuits build on ideas from classical digital logic. Understanding bits, Boolean functions, truth tables, and reversible computation makes the QPU gate model much easier to use.

## 1.1 Classical bits and logical state

A classical bit has one definite logical value:

$$b \in \{0, 1\}.$$

Physical computers may represent that value with voltage, charge, magnetic orientation, or another measurable quantity. Digital logic ignores most physical detail and works with the abstraction 0 or 1.

A wire carries a bit from one operation to another. A *gate* computes a function of one or more incoming bits. For example, NOT has one input and one output:

$$\text{NOT}(0) = 1, \quad \text{NOT}(1) = 0.$$

In QPU source, a token names the logical wire. On computational-basis states,  $X \text{ } -I \text{ } Q \text{ } -0 \text{ } Q$  has the same measured input/output relationship as classical NOT.

## 1.2 Boolean algebra

Boolean algebra describes functions whose values are 0 and 1. This guide uses:

| Operation   | Symbol             | Spoken form      | Meaning                             |
|-------------|--------------------|------------------|-------------------------------------|
| <b>NOT</b>  | $\neg A$           | not A            | 1 when A is 0; 0 when A is 1.       |
| <b>AND</b>  | $A \wedge B$       | A and B          | 1 only when both inputs are 1.      |
| <b>OR</b>   | $A \vee B$         | A or B           | 1 when either or both inputs are 1. |
| <b>XOR</b>  | $A \oplus B$       | A exclusive-or B | 1 when exactly one input is 1.      |
| <b>NAND</b> | $\neg(A \wedge B)$ | not-and          | The complement of AND.              |

For bits, XOR is addition modulo two:

$$0 \oplus 0 = 0, \quad 0 \oplus 1 = 1, \quad 1 \oplus 0 = 1, \quad 1 \oplus 1 = 0.$$

There is no carry in modulo-two addition. Ordinary binary addition handles the carry separately, which is why a half-adder produces both Sum and Carry.

## 1.3 Truth tables as complete finite specifications

A truth table lists a function's output for every possible input pattern. With  $n$  binary inputs there are  $2^n$  patterns. AND, OR, XOR, and NAND can be compared in one table:

| $A$ | $B$ | $A \wedge B$ | $A \vee B$ | $A \oplus B$ | $\neg(A \wedge B)$ |
|-----|-----|--------------|------------|--------------|--------------------|
| 0   | 0   | 0            | 0          | 0            | 1                  |
| 0   | 1   | 0            | 1          | 1            | 1                  |
| 1   | 0   | 0            | 1          | 1            | 1                  |
| 1   | 1   | 1            | 1          | 0            | 0                  |

For a deterministic combinational circuit, this table completely specifies the classical input/output behavior. It does not reveal the circuit's internal gate arrangement, propagation time, power consumption, or quantum phase.

## 1.4 Combinational logic and stored state

A *combinational* output depends only on current inputs. Adders and the Boolean gates above are combinational. A *sequential* circuit also depends on stored state from an earlier step, as in counters and latches.

QPU cycle suffixes can document successive stages, and INCREASECYCLE marks a workspace boundary, but the current compiled circuit is still one ordered gate sequence. Accepted `SAVE_STATE` and `LOAD_STATE` syntax does not yet implement a classical memory checkpoint. When modeling stateful behavior, expose previous state explicitly as `PARAMS` and return next state explicitly through `RETURNVALS`.

## 1.5 Why ordinary classical gates can lose information

A function is reversible only if each output identifies exactly one input. Classical NOT is reversible: seeing output 1 proves the input was 0. AND is not: output 0 could have come from 00, 01, or 10.

$$(A, B) \xrightarrow{\text{AND}} A \wedge B$$

maps four input patterns onto only two output values, discarding information. Quantum evolution through closed-system gates must be unitary and therefore reversible. A quantum/reversible AND preserves both inputs and mutates another target:

$$(A, B, t) \mapsto (A, B, t \oplus (A \wedge B)).$$

Knowing the output triple lets the transformation be run backward to recover the input triple.

## 1.6 Controls, targets, and workspace

These terms connect classical logic to QPU syntax:

**Control** A wire whose value decides whether a target operation occurs. In `CNOT -I A -O T`, A is the control.

**Target** The wire changed by an operation. T is the target in that CNOT.

**Workspace or ancilla** An extra wire introduced to hold an intermediate or output while preserving inputs. It is commonly initialized to  $|0\rangle$ .

**Garbage** Intermediate information left in workspace after the desired answer has been computed. Reversible algorithms often *uncompute* temporary gates to return garbage wires to zero.

**Uncomputation** Applying inverse operations in reverse order to clean workspace without erasing information irreversibly.

The QPU language calls logical wire names tokens. `CREATETOKEN` reserves workspace, `SET` prepares it, `-I` lists inputs/controls, and `-O` names targets.

## 1.7 Fan-out, copying, and the no-cloning boundary

Classical logic freely copies a bit onto several wires. If the source is a known basis bit and targets begin at zero, CNOT reproduces that basis value:

$$|x\rangle|0\rangle \xrightarrow{\text{CNOT}} |x\rangle|x\rangle, \quad x \in \{0, 1\}.$$

For a superposition  $\alpha|0\rangle + \beta|1\rangle$ , the same circuit produces

$$\alpha|00\rangle + \beta|11\rangle,$$

not

$$(\alpha|0\rangle + \beta|1\rangle) \otimes (\alpha|0\rangle + \beta|1\rangle).$$

The first state is generally entangled; the second would be two independent copies and contains cross terms. The no-cloning theorem forbids a universal gate that copies every unknown quantum state. QPU aliases similarly give two names to one wire; they do not create a physical copy.

## 1.8 Deterministic, probabilistic, and quantum models

| Model                          | State description                     | Transformation    | Observation                                                         |
|--------------------------------|---------------------------------------|-------------------|---------------------------------------------------------------------|
| <b>Deterministic classical</b> | One definite bit string               | Boolean function  | The definite output bits                                            |
| <b>Probabilistic classical</b> | Probabilities over bit strings        | Stochastic update | A sampled string                                                    |
| <b>Pure-state quantum</b>      | Complex amplitudes over basis strings | Unitary matrix    | A sampled string after magnitude-squared probabilities and collapse |

Negative and imaginary amplitudes are not negative or imaginary probabilities. They carry phase. Two computational paths can add constructively or cancel destructively before probabilities are calculated. This interference has no counterpart in an ordinary classical random-choice table.

## 1.9 Classical and QPU operation map

| Classical idea         | QPU operation  | Important difference                                                       |
|------------------------|----------------|----------------------------------------------------------------------------|
| <b>NOT</b>             | X or NOT       | Acts linearly on superpositions; X is unitary.                             |
| <b>XOR into target</b> | CNOT           | Preserves the control and can entangle.                                    |
| <b>AND into target</b> | CCNOT or AND   | Preserves controls; target is XOR-mutated.                                 |
| <b>Swap variables</b>  | SWAP           | Exchanges complete quantum wire states.                                    |
| <b>Random bit</b>      | H then MEASURE | H first creates coherent amplitudes, not an ordinary hidden random choice. |
| <b>Read a bit</b>      | MEASURE        | Sampling collapses a superposition and can disturb entanglement.           |
| <b>Initialize zero</b> | SET token 0p   | Compiles to state preparation/reset behavior.                              |
| <b>Function output</b> | RETURNVALS     | Declares an interface; does not itself copy or measure a wire.             |

## 2 Mathematical foundations

This chapter assumes no prior quantum mechanics. The essential ideas are numbers, vectors, probability, and transformations. Quantum notation can look unfamiliar, but each symbol has a concrete computational meaning.

### 2.1 Real numbers and complex numbers

The real numbers include familiar values such as  $-2$ ,  $0$ ,  $1/3$ , and  $\sqrt{2}$ . A real number can be placed on a one-dimensional number line. Quantum amplitudes require a second numerical direction, represented by the *imaginary unit*

$$i = \sqrt{-1}, \quad i^2 = -1.$$

There is no real number whose square is  $-1$ , so  $i$  extends the number system. A complex number has the form

$$z = a + bi,$$

where  $a$  is the real part and  $b$  is the imaginary part. Examples are  $3 + 2i$ ,  $-i$ , and  $1$  (which is  $1 + 0i$ ).

Complex arithmetic follows ordinary algebra together with  $i^2 = -1$ :

$$(a + bi) + (c + di) = (a + c) + (b + d)i,$$

$$(a + bi)(c + di) = (ac - bd) + (ad + bc)i.$$

For example,

$$(1 + i)^2 = 1 + 2i + i^2 = 2i.$$

## 2.2 The complex plane, magnitude, and conjugate

A complex number is a point or arrow in a plane: horizontal coordinate  $a$ , vertical coordinate  $b$ . Its magnitude is the arrow length,

$$|z| = \sqrt{a^2 + b^2}.$$

The complex conjugate reflects the arrow across the real axis:

$$z^* = a - bi.$$

Multiplying by the conjugate gives a nonnegative real number:

$$z^*z = (a - bi)(a + bi) = a^2 + b^2 = |z|^2.$$

Quantum measurement probabilities use  $|z|^2$ , not  $z$  itself. This guarantees real, nonnegative probabilities even though amplitudes may be complex.

## 2.3 Angles, radians, sine, and cosine

An arrow in the complex plane may be described by its length  $r$  and angle  $\theta$ :

$$a = r \cos \theta, \quad b = r \sin \theta.$$

Quantum formulas normally measure angles in radians. One full turn is  $2\pi$  radians, a half turn is  $\pi$ , and a quarter turn is  $\pi/2$ . Conversion is

$$\theta_{\text{radians}} = \theta_{\text{degrees}} \frac{\pi}{180}.$$

This is why `PHASE=90d` and `PHASE=pi/2` describe the same angle.

## 2.4 Euler's formula and Euler's identity

Euler's formula connects complex numbers to rotations:

$$e^{i\theta} = \cos \theta + i \sin \theta.$$

The value  $e^{i\theta}$  always has magnitude one. As  $\theta$  changes, it moves around the unit circle in the complex plane. Important cases are

$$e^{i0} = 1, \quad e^{i\pi/2} = i, \quad e^{i\pi} = -1, \quad e^{i2\pi} = 1.$$

Setting  $\theta = \pi$  gives Euler's identity,

$$e^{i\pi} + 1 = 0.$$

In a phase gate, multiplying an amplitude by  $e^{i\theta}$  rotates that amplitude without changing its magnitude. Its immediate measurement probability is unchanged, but its relationship to other amplitudes changes. Later gates can convert that relative phase into different probabilities.

## 2.5 Vectors and basis vectors

A column vector is an ordered list of numbers. The two basis vectors used for one qubit are

$$|0\rangle = \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \quad |1\rangle = \begin{bmatrix} 0 \\ 1 \end{bmatrix}.$$

The vertical bars and angle bracket form Dirac's *ket* notation. `0p` prepares  $|0\rangle$  and `1p` prepares  $|1\rangle$ . Any one-qubit pure state is a complex linear combination:

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle = \begin{bmatrix} \alpha \\ \beta \end{bmatrix}.$$

The complex numbers  $\alpha$  and  $\beta$  are *probability amplitudes*. They are not probabilities themselves.

## 2.6 Normalization and the Born rule

A valid state is normalized:

$$|\alpha|^2 + |\beta|^2 = 1.$$

The Born rule turns amplitudes into measurement probabilities:

$$P(0) = |\alpha|^2, \quad P(1) = |\beta|^2.$$

For the plus state,

$$|+\rangle = \frac{|0\rangle + |1\rangle}{\sqrt{2}},$$

both amplitudes are  $1/\sqrt{2}$ . Squaring their magnitudes gives  $P(0) = P(1) = 1/2$ . The factor  $1/\sqrt{2}$  is therefore required; writing  $|0\rangle + |1\rangle$  without normalization would produce total probability two.

## 2.7 Matrices as gates

A matrix is a rectangular array of numbers. A one-qubit gate is represented by a  $2 \times 2$  complex matrix. Matrix-vector multiplication computes the new amplitudes:

$$\begin{bmatrix} a & b \\ c & d \end{bmatrix} \begin{bmatrix} \alpha \\ \beta \end{bmatrix} = \begin{bmatrix} a\alpha + b\beta \\ c\alpha + d\beta \end{bmatrix}.$$

For example, the X gate exchanges the two amplitudes:

$$\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} \alpha \\ \beta \end{bmatrix} = \begin{bmatrix} \beta \\ \alpha \end{bmatrix}.$$

Applying gates in source order means multiplying by matrices in the same temporal order: if U acts first and V second, then

$$|\psi_{\text{final}}\rangle = V(U|\psi_{\text{initial}}\rangle) = VU|\psi_{\text{initial}}\rangle.$$

## 2.8 Unitarity and reversibility

Closed-system quantum gates must preserve total probability. Their matrices are *unitary*:

$$U^\dagger U = I.$$

Here  $U^\dagger$  means transpose the matrix and complex-conjugate every entry;  $I$  is the identity matrix. Unitarity guarantees an inverse,  $U^{-1} = U^\dagger$ . Quantum gates therefore cannot normally erase information. Classical functions such as AND are embedded reversibly by preserving inputs and XORing the result into a target:

$$|a, b, t\rangle \mapsto |a, b, t \oplus (a \wedge b)\rangle.$$

Measurement and reset are exceptions because they interact with classical outcomes or preparation rather than acting as closed unitary gates.

# 3 Qubits, phase, and the Bloch sphere

## 3.1 A qubit is not just an unknown bit

A classical bit is exactly 0 or 1. A qubit may be in any normalized superposition  $\alpha|0\rangle + \beta|1\rangle$ . This does not mean that the qubit secretly chose a classical value before measurement. Relative amplitudes can interfere, producing effects no ordinary random bit can reproduce.

### 3.2 Global phase and relative phase

Multiplying an entire state by a unit complex number does not change physical predictions:

$$|\psi\rangle \quad \text{and} \quad e^{i\gamma}|\psi\rangle$$

have the same measurement probabilities and are considered equivalent up to a *global phase*. Relative phase is different. The states

$$|+\rangle = \frac{|0\rangle + |1\rangle}{\sqrt{2}}, \quad |-\rangle = \frac{|0\rangle - |1\rangle}{\sqrt{2}}$$

give equal 0/1 probabilities if measured immediately, yet H sends  $|+\rangle \mapsto |0\rangle$  and  $|-\rangle \mapsto |1\rangle$ . Their relative sign is observable through interference.

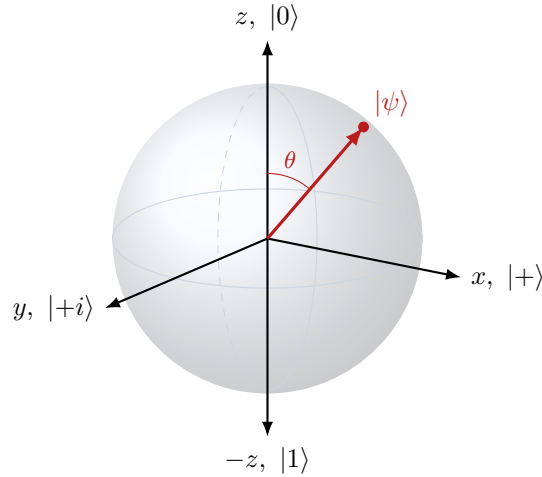
### 3.3 Bloch-sphere coordinates

After removing an unobservable global phase, every pure one-qubit state can be written with two real angles:

$$|\psi\rangle = \cos\left(\frac{\theta}{2}\right)|0\rangle + e^{i\phi}\sin\left(\frac{\theta}{2}\right)|1\rangle,$$

where  $0 \leq \theta \leq \pi$  and  $0 \leq \phi < 2\pi$ . These angles define a point on the unit Bloch sphere:

$$x = \sin\theta \cos\phi, \quad y = \sin\theta \sin\phi, \quad z = \cos\theta.$$



The north pole is  $|0\rangle$  and the south pole is  $|1\rangle$ . The equator contains equal-magnitude superpositions.  $|+\rangle$  points along  $+x$ ,  $|-\rangle = (|0\rangle - |1\rangle)/\sqrt{2}$  along  $-x$ , and  $|+i\rangle = (|0\rangle + i|1\rangle)/\sqrt{2}$  along  $+y$ .

### 3.4 How gates move a Bloch vector

Ignoring global phase, many one-qubit gates are rotations:

- X rotates the sphere by  $\pi$  around the x-axis, exchanging north and south.
- Y rotates by  $\pi$  around the y-axis.
- Z rotates by  $\pi$  around the z-axis, changing relative phase.
- S and T rotate around z by  $\pi/2$  and  $\pi/4$ .
- PHASE rotates around z by its supplied angle  $\theta$ .
- H rotates the coordinate description so the z basis and x basis are exchanged:  $|0\rangle \leftrightarrow |+\rangle$  and  $|1\rangle \leftrightarrow |-\rangle$ .

The sphere is a teaching model for one pure qubit. It cannot display the full state of entangled multi-qubit systems as one ordinary three-dimensional arrow.

## 4 Multiple qubits and entanglement

### 4.1 Tensor products

Two classical bits have four possibilities; two qubits have four basis vectors:

$$|00\rangle, |01\rangle, |10\rangle, |11\rangle.$$

Independent systems combine with the tensor product  $\otimes$ . If  $|A\rangle = [a_0, a_1]^T$  and  $|B\rangle = [b_0, b_1]^T$ , then

$$|A\rangle \otimes |B\rangle = \begin{bmatrix} a_0 b_0 \\ a_0 b_1 \\ a_1 b_0 \\ a_1 b_1 \end{bmatrix}.$$

For  $n$  qubits, the simulator stores  $2^n$  amplitudes:

$$|\Psi\rangle = \sum_{x=0}^{2^n-1} a_x |x\rangle, \quad \sum_x |a_x|^2 = 1.$$

This exponential growth is why state-vector simulation becomes expensive as qubit count increases.

### 4.2 Entanglement

Some multi-qubit states cannot be factored into independent one-qubit states. The Bell state

$$|\Phi^+\rangle = \frac{|00\rangle + |11\rangle}{\sqrt{2}}$$

is entangled. Each qubit measured alone is equally likely to be 0 or 1, but their outcomes are perfectly correlated. It can be prepared from  $|00\rangle$ :

```
H -I A -0 A
CNOT -I A -0 B
```

The first gate creates a superposition; the controlled gate correlates B with A. Entanglement is not a classical alias. It is a property of the joint amplitude vector.

### 4.3 Controlled operations

A controlled gate applies a target operation only on basis components where its control condition is true. CNOT applies X when its control is one. If the control is in superposition, linearity applies the operation to one branch and not the other, which can create entanglement:

$$\frac{|0\rangle + |1\rangle}{\sqrt{2}} \otimes |0\rangle \xrightarrow{\text{CNOT}} \frac{|00\rangle + |11\rangle}{\sqrt{2}}.$$

CCNOT requires both controls to be one. The system's OR and XOR derived gates use “any active” and “odd parity” predicates respectively.

## 5 Measurement, probability, and truth tables

### 5.1 Projective measurement

For a measured qubit  $q$  in a multi-qubit state,

$$P(q=1) = \sum_{x: x_q=1} |a_x|^2, \quad P(q=0) = 1 - P(q=1).$$

The simulator samples one outcome. Amplitudes incompatible with that outcome become zero; compatible amplitudes are divided by the square root of the kept probability so the state is normalized again. Measurement is destructive in the sense that the pre-measurement superposition is generally unavailable after collapse.

## 5.2 Single runs and repeated runs

One simulation run produces one measurement sample. To estimate a probability, run the same preparation and circuit many times. If a plus state is measured 1 approximately 500 times in 1000 shots, the observed frequency 500/1000 estimates the theoretical probability 1/2. More shots reduce ordinary sampling noise but never reveal a complex amplitude directly.

## 5.3 Truth tables versus quantum amplitudes

A classical truth table specifies output bits for listed input bits. It is a good contract for arithmetic and reversible Boolean circuits operating on computational-basis states. It cannot fully describe phase, interference, entanglement, or a probability distribution.

The QPU truth-table runner prepares each input row, executes the circuit, measures declared RETURNVALS, and compares those measured bits. In companion `.qpuio` tables, cells are `0p`, `1p`, or `sp`. On an expected output, `sp` means “either measured bit is acceptable.” Use state equations and repeated experiments in addition to tables when a circuit’s purpose is genuinely quantum.

## 5.4 A complete interference example

Starting in zero, two Hadamards restore zero:

$$|0\rangle \xrightarrow{H} |+\rangle \xrightarrow{H} |0\rangle.$$

Inserting Z changes the relative phase and changes the final answer:

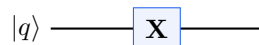
$$|0\rangle \xrightarrow{H} |+\rangle \xrightarrow{Z} |-\rangle \xrightarrow{H} |1\rangle.$$

```
MAIN-PROCESS Interference
CREATETOKEN -I Q
SET Q 0p
H -I Q -0 Q
Z -I Q -0 Q
H -I Q -0 Q
MEASURE -I Q
RETURNVALS Q
```

The middle Z does not change an immediate 0/1 probability, yet interference at the final H makes its effect deterministic. This is the central reason the system documents matrices and phase as well as truth tables.

# 6 Single-qubit primitive gates

## 6.1 Pauli X (X)



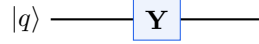
$$X = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}, \quad X|0\rangle = |1\rangle, \quad X|1\rangle = |0\rangle, \quad X(\alpha|0\rangle + \beta|1\rangle) = \beta|0\rangle + \alpha|1\rangle.$$

X is the quantum bit flip. On the Bloch sphere it is a half-turn around the x-axis. It exchanges the north and south poles and is self-inverse:  $X^2 = I$ . Unlike a classical NOT implementation, X remains meaningful on a superposition because it exchanges amplitudes linearly without measuring.

**X** -I Q:0 -0 Q:0

| Input | Classical comparison | Output |
|-------|----------------------|--------|
| 0     | NOT 0                | 1      |
| 1     | NOT 1                | 0      |

## 6.2 Pauli Y (Y)



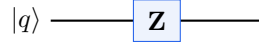
$$Y = \begin{bmatrix} 0 & -i \\ i & 0 \end{bmatrix}, \quad Y|0\rangle = i|1\rangle, \quad Y|1\rangle = -i|0\rangle.$$

It flips the basis value like X but adds phases that become observable through later interference. Geometrically it is a half-turn around the y-axis and is self-inverse,  $Y^2 = I$ . The factors  $i$  and  $-i$  are global phases for either basis state by itself, but on components of a larger superposition they may become relative phases.

**Y** -I Q -0 Q

| Input | Measured bit | Quantum output |
|-------|--------------|----------------|
| 0     | 1            | $i 1\rangle$   |
| 1     | 0            | $-i 0\rangle$  |

## 6.3 Pauli Z (Z)



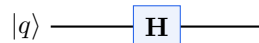
$$Z = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}, \quad Z|0\rangle = |0\rangle, \quad Z|1\rangle = -|1\rangle.$$

Z does not change a computational-basis measurement, but it changes relative phase. For example,  $Z|+\rangle = |-\rangle = (|0\rangle - |1\rangle)/\sqrt{2}$ . On the Bloch sphere it is a half-turn around z. It is self-inverse and is the simplest example showing why quantum state cannot be reduced to a table of immediate measurement probabilities.

**Z** -I Q -0 Q

| Input | Classical bit | Quantum output |
|-------|---------------|----------------|
| 0     | 0             | $ 0\rangle$    |
| 1     | 1             | $- 1\rangle$   |

## 6.4 Hadamard (H)



$$H = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}, \quad H|0\rangle = |+\rangle, \quad H|1\rangle = |-\rangle.$$

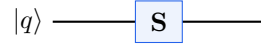
Hadamard changes between the computational z basis and the plus/minus x basis. It creates equal magnitudes from a basis state, but the sign of the second amplitude depends on the input. It is also self-inverse: applying H twice returns the original state. H is commonly used before controlled gates to create entanglement and after phase gates to reveal interference.

```
SET Q 0p
H -I Q -0 Q
MEASURE -I Q
```

| Input | Measurement distribution | State before measurement           |
|-------|--------------------------|------------------------------------|
| 0     | 50% 0, 50% 1             | $( 0\rangle +  1\rangle)/\sqrt{2}$ |
| 1     | 50% 0, 50% 1             | $( 0\rangle -  1\rangle)/\sqrt{2}$ |

Equal classical probabilities hide the relative minus sign responsible for interference.

## 6.5 S phase (S)



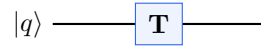
$$S = P(\pi/2) = \begin{bmatrix} 1 & 0 \\ 0 & i \end{bmatrix}, \quad S|1\rangle = i|1\rangle, \quad S^2 = Z.$$

S is a quarter-turn around the Bloch sphere's z-axis. It changes  $|+\rangle$  into  $(|0\rangle + i|1\rangle)/\sqrt{2}$  while leaving basis measurement probabilities unchanged. Its inverse is  $S^\dagger = P(-\pi/2)$ , not S itself.

```
S -I Q -0 Q
```

| Input | Measured bit | Quantum output |
|-------|--------------|----------------|
| 0     | 0            | $ 0\rangle$    |
| 1     | 1            | $i 1\rangle$   |

## 6.6 T phase (T)



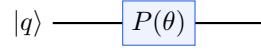
$$T = P(\pi/4) = \begin{bmatrix} 1 & 0 \\ 0 & e^{i\pi/4} \end{bmatrix}, \quad T^2 = S, \quad T^4 = Z.$$

T is an eighth-turn around z. Together with gates such as H and CNOT, it is an important non-Clifford resource for expressing general quantum computations. Its inverse is  $T^\dagger = P(-\pi/4)$ ; the current BT spelling does not synthesize that inverse, so use `BPHASE=pi/4` when the negative phase is required.

```
T -I Q -0 Q
```

| Input | Measured bit | Quantum output        |
|-------|--------------|-----------------------|
| 0     | 0            | $ 0\rangle$           |
| 1     | 1            | $e^{i\pi/4} 1\rangle$ |

## 6.7 Arbitrary phase (PHASE)



$$P(\theta) = \begin{bmatrix} 1 & 0 \\ 0 & e^{i\theta} \end{bmatrix}, \quad \alpha|0\rangle + \beta|1\rangle \mapsto \alpha|0\rangle + \beta e^{i\theta}|1\rangle.$$

PHASE is the general form behind S, T, and Z. It preserves  $|\alpha|^2$  and  $|\beta|^2$  because  $|e^{i\theta}| = 1$ , yet rotates the relative phase by  $\theta$ . On the Bloch sphere this is a z-axis rotation up to global phase. A later H or another basis-changing gate is normally needed to turn that phase difference into changed computational-basis probabilities. The angle is embedded in the opcode:

```
PHASE=1.5708 -I Q -0 Q      # radians
PHASE=3/2 -I Q -0 Q        # rational radians
PHASE=90d -I Q -0 Q        # degrees
PHASE=45/2d -I Q -0 Q      # rational degrees
PHASE=pi -I Q -0 Q
PHASE=-11pi/6 -I Q -0 Q
PHASE=2*pi/3 -I Q -0 Q
BPHASE=pi/4 -I Q -0 Q      # P(-pi/4)
```

Accepted numeric atoms include signed integers, decimals, leading-dot decimals, and one optional rational denominator. Pi notation accepts an optional sign, an integer coefficient, an optional \*, and an optional positive integer denominator. A zero denominator and malformed angle raise an error.

| Input | Measured bit | Quantum output         |
|-------|--------------|------------------------|
| 0     | 0            | $ 0\rangle$            |
| 1     | 1            | $e^{i\theta} 1\rangle$ |

## 6.8 Backward-prefix syntax

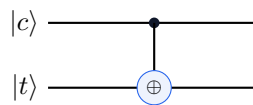
The QPU language recognizes a leading B before a supported primitive gate name, for example BX, BH, BCNOT, or BSWAP, and sets `reverse=true`. (Compiler-internal RESET remains unavailable in source.) Only BPHASE=angle currently changes emitted behavior: the compiler negates the phase angle. For every other gate the flag is metadata only. Emitting the ordinary gate still gives the mathematical inverse for these self-inverse gates:

X, Y, Z, H, CNOT, CCNOT, CZ, CY, SWAP.

In contrast, BS and BT do not synthesize  $S^\dagger$  or  $T^\dagger$ . Do not rely on B-prefixed forms for inverse execution unless the gate is self-inverse or is BPHASE.

# 7 Controlled and two-qubit primitive gates

## 7.1 Controlled NOT (CNOT)



$$|c, t\rangle \mapsto |c, t \oplus c\rangle.$$

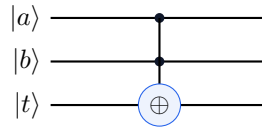
CNOT leaves the control unchanged. When the control is zero, the target is unchanged; when the control is one, X acts on the target. Because the map is a permutation of basis states, it is reversible and self-inverse. A superposition control can entangle the wires, so “classical XOR” describes only its basis-state action.

**CNOT** -I Control -O Target

| $c$ | $t$ | $c'$ | $t'$ |
|-----|-----|------|------|
| 0   | 0   | 0    | 0    |
| 0   | 1   | 0    | 1    |
| 1   | 0   | 1    | 1    |
| 1   | 1   | 1    | 0    |

CNOT is a reversible embedding of XOR because it preserves the control.

## 7.2 Doubly controlled NOT / Toffoli (CCNOT)



$$|a, b, t\rangle \mapsto |a, b, t \oplus (a \wedge b)\rangle.$$

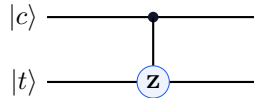
CCNOT, also called Toffoli, applies X only when both controls are one. It is a reversible universal building block for classical computation embedded in a quantum circuit. The target's previous value matters: zero computes AND, while one computes NAND. The gate is self-inverse.

**CCNOT** -I A B -O Target

| $a$ | $b$ | $t$ | $a'$ | $b'$ | $t'$ |
|-----|-----|-----|------|------|------|
| 0   | 0   | 0   | 0    | 0    | 0    |
| 0   | 0   | 1   | 0    | 0    | 1    |
| 0   | 1   | 0   | 0    | 1    | 0    |
| 0   | 1   | 1   | 0    | 1    | 1    |
| 1   | 0   | 0   | 1    | 0    | 0    |
| 1   | 0   | 1   | 1    | 0    | 1    |
| 1   | 1   | 0   | 1    | 1    | 1    |
| 1   | 1   | 1   | 1    | 1    | 0    |

With a target initialized to zero, Toffoli computes classical AND reversibly.

## 7.3 Controlled Z (CZ)



$$\text{CZ}|c, t\rangle = (-1)^{ct}|c, t\rangle, \quad \text{CZ} = \text{diag}(1, 1, 1, -1).$$

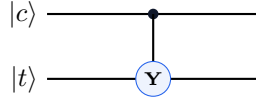
CZ adds a minus sign only to the  $|11\rangle$  amplitude. No computational-basis bit changes, but superpositions can acquire observable relative phase. CZ is symmetric between its two wires, even though QPU syntax writes one as the input control and one as the output target. It is self-inverse.

**CZ** -I Control -O Target

| Input | Measured output | Amplitude multiplier | State         |
|-------|-----------------|----------------------|---------------|
| 00    | 00              | +1                   | $ 00\rangle$  |
| 01    | 01              | +1                   | $ 01\rangle$  |
| 10    | 10              | +1                   | $ 10\rangle$  |
| 11    | 11              | -1                   | $- 11\rangle$ |

The classical columns are unchanged; the quantum phase is the operation.

## 7.4 Controlled Y (CY)



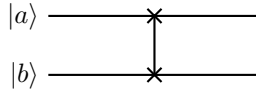
$$|0, t\rangle \mapsto |0, t\rangle, \quad |1, 0\rangle \mapsto i|1, 1\rangle, \quad |1, 1\rangle \mapsto -i|1, 0\rangle.$$

CY applies the Pauli Y rotation to the target only when the control is one. Its basis-bit pattern resembles CNOT, but the  $\pm i$  factors distinguish their behavior on superpositions. Since  $Y^2 = I$ , CY is self-inverse.

**CY** -I Control -0 Target

| $c$ | $t$ | $c'$ | $t'$ | Quantum phase |
|-----|-----|------|------|---------------|
| 0   | 0   | 0    | 0    | +1            |
| 0   | 1   | 0    | 1    | +1            |
| 1   | 0   | 1    | 1    | $i$           |
| 1   | 1   | 1    | 0    | $-i$          |

## 7.5 Swap (SWAP)



$$\text{SWAP}|a, b\rangle = |b, a\rangle.$$

SWAP exchanges the complete quantum states of two wires, including amplitudes and any correlations with the rest of the register. It does not copy either state. Three CNOTs can decompose SWAP, but the QPU system provides it as one primitive instruction. Applying SWAP twice restores the original ordering.

**SWAP** -I A B -0 A B

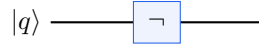
The two output tokens are mandatory and must resolve to the same two wires, in the same order, as the inputs.

| $a$ | $b$ | $a'$ | $b'$ |
|-----|-----|------|------|
| 0   | 0   | 0    | 0    |
| 0   | 1   | 1    | 0    |
| 1   | 0   | 0    | 1    |
| 1   | 1   | 1    | 1    |

## 8 Derived logical gates

Derived gates are QPU instructions implemented as reversible target mutations. Inputs are preserved. If target  $t$  starts in zero, AND/OR/XOR write the familiar classical function. If it does not, the true behavior is the XOR embedding shown below.

### 8.1 Logical NOT (NOT)



$$t' = t \oplus 1.$$

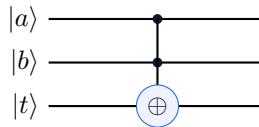
NOT is the derived logical spelling of a bit flip. In the current simulator it uses the same X matrix on the output target. The required input establishes the one-input instruction shape, but a distinct input is not copied into the output; use the same token on both sides.

**NOT** -I Target -O Target

| $t$ | $t'$ |
|-----|------|
| 0   | 1    |
| 1   | 0    |

The current gate implementation mutates the output target. Use the same token for input and output; a distinct input token is not copied.

### 8.2 Reversible AND (AND)



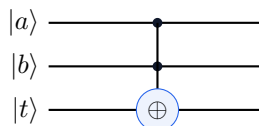
$$t' = t \oplus (a \wedge b).$$

AND flips the target exactly on the basis components where both controls are one. This is Toffoli-style behavior. It remains linear on superpositions and may entangle the target with its controls. Calling it AND is convenient only when the target is deliberately initialized.

**SET** Out Op  
**AND** -I A B -O Out

| $a$ | $b$ | $t = 0$ | $t' = a \wedge b$ |
|-----|-----|---------|-------------------|
| 0   | 0   | 0       | 0                 |
| 0   | 1   | 0       | 0                 |
| 1   | 0   | 0       | 0                 |
| 1   | 1   | 0       | 1                 |

### 8.3 Reversible NAND (NAND)



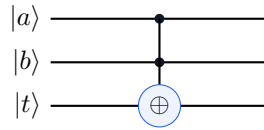
$$t' = t \oplus (a \wedge b) \oplus 1.$$

NAND first performs the all-controls-active mutation and then inverts the target. With target zero it produces the familiar NAND column. With target one it instead produces AND, illustrating why reversible target state must always be included in the full mathematical definition.

**SET** Out Op  
**NAND** -I A B -0 Out

| $a$ | $b$ | $t = 0$ | $t' = \neg(a \wedge b)$ |
|-----|-----|---------|-------------------------|
| 0   | 0   | 0       | 1                       |
| 0   | 1   | 0       | 1                       |
| 1   | 0   | 0       | 1                       |
| 1   | 1   | 0       | 0                       |

## 8.4 Reversible OR (OR)



$$t' = t \oplus (a \vee b).$$

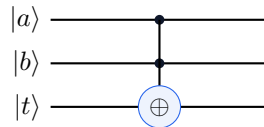
OR uses an “any control is one” predicate. If either or both controls are active, the target is flipped. This direct derived operation is equivalent on basis states to combining XOR and AND through  $a \vee b = a \oplus b \oplus (a \wedge b)$ .

**SET** Out Op  
**OR** -I A B -0 Out

| $a$ | $b$ | $t = 0$ | $t' = a \vee b$ |
|-----|-----|---------|-----------------|
| 0   | 0   | 0       | 0               |
| 0   | 1   | 0       | 1               |
| 1   | 0   | 0       | 1               |
| 1   | 1   | 0       | 1               |

For bits,  $a \vee b = a \oplus b \oplus (a \wedge b)$ .

## 8.5 Reversible XOR (XOR)



$$t' = t \oplus a \oplus b.$$

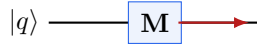
XOR flips the target when an odd number of controls is one. With two inputs, this is the ordinary exclusive-or predicate. The parity definition explains why the 11 row does not flip: two active controls have even parity.

**SET** Out Op  
**XOR** -I A B -0 Out

| $a$ | $b$ | $t = 0$ | $t' = a \oplus b$ |
|-----|-----|---------|-------------------|
| 0   | 0   | 0       | 0                 |
| 0   | 1   | 0       | 1                 |
| 1   | 0   | 0       | 1                 |
| 1   | 1   | 0       | 0                 |

## 9 Measurement gate

### 9.1 Measurement (MEASURE)



```
MEASURE -I Q      # exactly one named wire
MEASURE           # every wire known at this point
```

MEASURE takes no -O. Repeated one-wire commands are useful when only declared returns should become classical. A bare command emits one measurement operation per currently allocated wire.

| State                              | Outcome probabilities                 | Collapsed state            |
|------------------------------------|---------------------------------------|----------------------------|
| $ 0\rangle$                        | $P(0) = 1$                            | $ 0\rangle$                |
| $ 1\rangle$                        | $P(1) = 1$                            | $ 1\rangle$                |
| $\alpha 0\rangle + \beta 1\rangle$ | $P(0) =  \alpha ^2, P(1) =  \beta ^2$ | $ 0\rangle$ or $ 1\rangle$ |

## 10 Glossary

| Term                       | Meaning                                                                                                                                                                                            |
|----------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Amplitude</b>           | A complex coefficient of a basis state. Its squared magnitude contributes to measurement probability.                                                                                              |
| <b>Ancilla</b>             | An auxiliary qubit or workspace wire, usually prepared in a known state, used to hold intermediate information in a reversible computation.                                                        |
| <b>Basis state</b>         | A state used as a coordinate direction. The computational basis for one qubit is $ 0\rangle,  1\rangle$ .                                                                                          |
| <b>Bloch sphere</b>        | A geometric representation of pure one-qubit states after ignoring global phase. North/south are $ 0\rangle/ 1\rangle$ ; longitude represents relative phase.                                      |
| <b>Boolean algebra</b>     | The mathematics of 0/1 values and operations such as NOT, AND, OR, and XOR.                                                                                                                        |
| <b>Child process</b>       | A separately named protocol expanded into a parent through RUNCHILD or CALL.                                                                                                                       |
| <b>Combinational logic</b> | A classical circuit whose output depends only on current inputs, not stored history.                                                                                                               |
| <b>Control</b>             | A wire whose basis condition decides whether a controlled operation acts on its target. The control itself is preserved by the standard controlled gates.                                          |
| <b>Entanglement</b>        | A joint multi-qubit state that cannot be written as a tensor product of independent qubit states.                                                                                                  |
| <b>Fan-out</b>             | Sending one classical value to multiple destinations. Unknown quantum states cannot be universally copied; CNOT copies only known basis information onto a zero target and otherwise may entangle. |
| <b>Gate</b>                | A state transformation. Closed-system quantum gates are unitary; measurement and reset have different semantics.                                                                                   |

| Term                  | Meaning                                                                                                                              |
|-----------------------|--------------------------------------------------------------------------------------------------------------------------------------|
| <b>Garbage</b>        | Reversible-workspace information no longer wanted as output but still present because it cannot simply be erased.                    |
| <b>Global phase</b>   | A unit complex multiplier applied to the whole state. It does not change physical measurement predictions.                           |
| <b>Ket</b>            | Dirac notation $ \psi\rangle$ for a state column vector.                                                                             |
| <b>Measurement</b>    | Conversion of quantum amplitudes into a sampled classical result, with state collapse and renormalization.                           |
| <b>Phase</b>          | The angular part of a complex amplitude. Relative phase controls interference even when immediate basis probabilities are unchanged. |
| <b>Process</b>        | A named QPU protocol body with ordered PARAMS and RETURNVALS.                                                                        |
| <b>Protocol</b>       | Plain-text QPU instructions describing preparation, gates, composition, measurement, and returned references.                        |
| <b>Qubit</b>          | A normalized two-component complex state $\alpha 0\rangle + \beta 1\rangle$ , not merely an unknown classical bit.                   |
| <b>Register</b>       | One or more logical wires considered together. Current JOIN/SPLIT syntax is accepted but does not yet lower to register operations.  |
| <b>Relative phase</b> | The phase difference between amplitudes. Unlike global phase, it can change later interference and measurement probabilities.        |
| <b>Superposition</b>  | A coherent linear combination of basis states. <code>sp</code> prepares the equal plus superposition.                                |
| <b>Target</b>         | The wire whose amplitudes a gate mutates; normally the first <code>-0</code> reference.                                              |
| <b>Tensor product</b> | The operation $\otimes$ that combines state spaces. Two one-qubit vectors produce a four-amplitude two-qubit vector.                 |
| <b>Token</b>          | A source-level logical name resolved to a qubit wire or alias during compilation.                                                    |
| <b>Truth table</b>    | Listed input/output expectations for measured RETURNVALS. It describes basis behavior but not arbitrary phase or entanglement.       |
| <b>Uncomputation</b>  | Running suitable inverse gates in reverse order to return temporary workspace to a known state without irreversible erasure.         |
| <b>Unitary</b>        | A matrix satisfying $U^\dagger U = I$ , preserving normalization and having a reversible inverse.                                    |